Machine 机器

Learning 学习

Applications 应用

for Health 用于健康

COMP90089

Instructor Info - 授课教师信息 -

Prof Daniel Capurro, PhD and Dr. Vlada Rozova

丹尼尔·卡普罗教授（Prof Daniel Capurro），博士（PhD）和弗拉达·罗佐娃博士（Dr. Vlada Rozova）

Office Hrs: By appointment

办公时间：预约制

700 Swanston St, Level 4 (Melbourne Connect

墨尔本连接中心，斯旺斯顿街 700 号，4 层 (Melbourne Connect dcapurro@unimelb.edu.au, vlada.rozova@unimelb.edu.au

dcapurro@unimelb.edu.au、vlada.rozova@unimelb.edu.au

Course Info 课程信息

Tuesdays 周二

3:00am-5:00pm 下午 3:00-5:00

Q230 Theatre - Kwong Lee Dow Building - Level 2

Q230 讲堂 - Kwong Lee Dow 大楼 - 2 层

Tutor Info 导师信息

Roben Delos Reyes 罗本·德洛斯·雷耶斯
r.delosreyes@unimelb.edu.au

Jemima Kang
jemima.kang.1@unimelb.edu.au

Jie Huang 黄杰
jie.huang.4@unimelb.edu.au

Overview 概述

Artificial Intelligence (AI) is an ever growing field that holds the promise to revolutionise the way we develop drugs, treat and manage patients. In particular, Machine Learning has become an important tool to make sense of clinical data that is routinely collected and stored as electronic health records to enable personalised medicine.

人工智能（AI）是一个不断发展的领域，有望彻底改变我们开发药物、治疗和管理病患的方式。特别是，机器学习已成为理解常规收集并以电子健康记录形式存储的临床数据的重要工具，从而实现个性化医疗。

This subject aims to introduce students to different Machine Learning applications in health, using different clinical data sources and computational techniques, discussing their idiosyncrasies and the main challenges in the area.

本课程旨在向学生介绍在健康领域中不同的机器学习应用，使用不同的临床数据来源和计算技术，讨论它们的特殊性以及该领域的主要挑战。

Material 材质

Required Texts 必需文本

There are no required text books on this subject. Any required journal articles, book chapters, or other reading material will be provided on Canvas. Videos will also be provided on Canvas.

本课程没有指定教材。任何必需的期刊文章、书籍章节或其他阅读材料将通过 Canvas 提供。视频也将通过 Canvas 提供。

Assessment 评估

Note that you must achieve at least a 50% pass in each of the individual and group assignments in order to pass the subject.

请注意，您必须在个人作业和小组作业中每项至少达到 50%的通过分数，才能通过本课程。

20%
Regular, in-class (weekly) quizzes - Equivalent to 500 words in total, from week 2 to week 11

每周课堂小测 - 总计相当于 500 字，覆盖第 2 周到第 11 周

30%
Two individual assignments - Equivalent to 2000 words (2×1000 words). These will be individual programming assignments about one of the Machine Learning applications in health presented during the subject. Due in Week 4 & 10.

两个个人作业 - 总计相当于 2000 字（2×1000 字）。这些将是关于课程中呈现的某一健康领域机器学习应用的个人编程作业。第 4 周和第 10 周截止。

AI in health project proposal (group activity, 4 students, oral presentation) - Equivalent to 500 words per group. Due in Week 8.

健康领域的人工智能项目提案（小组活动，4 名学生，口头报告） - 每组相当于 500 字。第 8 周截止。

AI in health project written report (20%) + oral presentation (20%)(group activity, 4 students) - Equivalent to 2000 words per group. Due at the start of the exam period.

健康领域的人工智能项目书面报告（20%）+ 口头报告（20%）（小组活动，4 名学生） - 每组相当于 2000 字。在考试期开始时截止。

Every assignment will be checked for originality and the expectation is that every work will be:

每一份作业都将进行原创性检查，期望每份作品都应为：

- Original work done by the author(s) of the assignment during this subject, during this semester;

  由作业作者在本课程、本学期内独立完成的原创作品；

- Appropriately referenced (meaning every claim needs a reference; references will be checked). You can pick the citation style but please be consistent.

  适当引用（即每一项论断都需有出处；引用将被核查）。你可以选择引用格式，但请保持一致。

- Generative AI can be used for brainstorming, planning, proofreading and editing (as described here) but should not be used for drafting text or content generation. Its use should be disclosed as per the University of Melbourne's current policy. Every written/code assignment should have a statement by the student or group of students stating:

  可以使用生成式人工智能进行头脑风暴、规划、校对和编辑（如此处所述），但不应用于起草文本或生成内容。其使用应按照墨尔本大学的现行政策进行披露。每份书面/代码作业都应由学生或学生组提供一份声明，说明：

  - Whether a generative AI was used

    是否使用了生成式人工智能

  - Which tool was used 使用了哪种工具

- The purpose it was used for (brainstorming, planning, proofreading, or editing)

  用于的目的（头脑风暴、计划、校对或编辑）

- A declaration that it was not used for drafting text/code or content generation.

  声明未用于起草文本/代码或生成内容。

In case the work does not meet these expectations, the assignment will penalised with a 30% discount on the mark. A late penalty of 5% per unjustified day of delay also applies.

如果工作不符合这些期望，作业将按 30% 的比例扣分。对于无正当理由的迟交，每延迟一天还将按 5% 收取罚分。

Curving of grades is at the discretion of the course instructor.

成绩的调整由课程教师自行裁量。

FAQs 常见问题

? Can I pick my own group for the project?

? 我可以为项目挑选自己的小组吗?

(B) No, your groups will be assigned by the instructor so they are diverse enough to enhance peer learning.

不可以，教师会分配小组，以确保小组具有足够的多样性，从而促进同伴学习。

? Can I pick my own topic for the project?

? 我可以为项目挑选自己的课题吗?

(B) Yes, but you need to agree as a group what your topic will be.

是的，但你们需要作为一个小组就主题达成一致。

? Can I do my project individually?

? 我可以单独完成我的项目吗?

(B) No 否

? What can I do if my group is not working for me?

? 如果我的小组对我不起作用，我能做什么?

(B) Refer to the group contract you signed on week 1 and please let the instructor know as soon as possible.

请参考你在第 1 周签署的小组合同，并尽快告知授课教师。

Learning Objectives 学习目标

On completion of this subject, students should be able to:

完成本课程后，学生应能够：

- Describe the application of AI concepts in the context of health problems

  描述人工智能概念在健康问题背景下的应用

- Demonstrate familiarity with challenges associated with health data and data modelling

  展示对与健康数据和数据建模相关挑战的熟悉程度

- Design, implement & evaluate an AI system addressing a healthcare problem

  设计、实现并评估一个解决医疗健康问题的人工智能系统

- Critically assess Artificial Intelligence applications for health

  批判性地评估人工智能在健康领域的应用

Generic Skills: 通用技能：

- Develop critical thinking and analytical skills

  培养批判性思维和分析能力

- Improve problem-solving skills

  提高解决问题的能力

- Enhance programming skills

  增强编程技能

- Improve communication skills

  提升沟通能力

- Develop skills to enable problem identification, solution formulation for health applications

  培养能够识别问题并为健康应用制定解决方案的技能

Subject Communication 课程交流

Announcements will be posted on Ed Discussion, please make sure you review the platform and your student email address.

公告将发布在 Ed Discussion 上，请务必查看该平台并检查你的学生邮箱地址。

If you need to communicate with the instructors or tutors, please use Ed Discussion. If you want to send an email, use the following text in the email's subject: [COMP90089 inquiry].

如果你需要与授课教师或助教沟通，请使用 Ed Discussion。若要发送电子邮件，请在邮件主题中使用以下文本：[COMP90089 inquiry]。

If you have questions about the subject, assessments, etc, please post them at the course's forum so everyone can see the question and the answers. This forum is also good for establishing student-to-student interactions.

如果你对课程、评估等有疑问，请将问题发布在课程论坛，以便所有人都能看到问题及其答案。该论坛也有助于建立学生之间的互动。

Diversity and Inclusivity Statement

多样性与包容性声明

I consider this course to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.

我认为这门课程是一个你将受到尊重的地方，我欢迎所有年龄、背景、信仰、种族、性别、性别认同、性别表达、国籍、宗教隶属、性取向、能力及其他可见或不可见差异的个体。班上每一位成员都应为为其他所有成员创造一个尊重、友好且包容的环境做出贡献。

Student Support Services

学生支持服务

The University offers many support services for students.

大学为学生提供多种支持服务。

Some of the key services are listed below (links are clickable):

以下列出了一些主要服务（链接可点击）：

- Library 图书馆
- Academic Skills 学术技能
- Student Equity and Disability Support

  学生公平与残障支持
- Student wellbeing 学生福祉
- Special consideration processes

  特殊照顾程序
- Stop 1 Support 第一站支持
- Career support 职业支持

Academic Integrity 学术诚信

The University has a Student Academic Integrity Policy. Students are expected to be familiar with the code and to understand that every submitted assignment is a product of their own original work and faithfully represents the amount of effort and time invested in producing it. Every use of external work should be adequately referenced. Violations of the policy will be handled according to the procedures described in the aforementioned policy.

大学有一项学生学术诚信政策。学生应熟悉该守则，并理解每份提交的作业都应为其原创作品，且真实反映为完成该作业所投入的时间和努力。所有使用的外部资料都应给予适当引用。违反该政策的行为将按照前述政策中所述的程序处理。

You can also check the Academic Integrity Website. Additional information is available in the Academic Integrity Module inside the course's Canvas site.

你也可以查看学术诚信网站。课程 Canvas 网站内的学术诚信模块中提供了更多信息。

Class Schedule 课程安排

MODULE 1: Introduction 模块 1：导论		
Week-1 28/07/26 第 1 周 26/07/28	Lecture #1: Course Introduction & Housekeeping 讲座 #1：课程介绍与事务安排	Prof Daniel Capurro 丹尼尔·卡普罗罗 教授
	Lecture #2: Biomedical Informatics & the Learning Healthcare System 讲座 #2：生物医学信息学与学习型医疗系统	Prof Daniel Capurro 丹尼尔·卡普罗罗 教授
Week-2 04/08/26 第 2 周 26/08/04	Lecture #3: Data Sources 讲座 #3：数据来源	Prof Daniel Capurro 丹尼尔·卡普罗教授
	Lecture #4: Clinical Data Types 第 4 讲：临床数据类型	Dr Vlada Rozova 弗拉达·罗佐娃博士
Week-3 11/08/26 第 3 周 2026/08/11	Lecture #5: Privacy, De-identification, Ethics, and Governance 第 5 讲：隐私、去识别、伦理与治理	Dr. Vlada Rozova 弗拉达·罗佐娃博士
	Lecture #6: Deep Dive into Clinical Data 第 6 讲：临床数据深入探讨	Prof Daniel Capurro 丹尼尔·卡普罗教授
MODULE 2: Why is healthcare AI unique? 模块 2：为什么医疗保健中的人工智能是独特的?		
Week-4 18/08/26 第 4 周 2018/08/26	Lecture #7: Digital Phenotyping 讲座 #7：数字表型	Prof Daniel Capurro 丹尼尔·卡普罗教授
Week-5 25/08/26 第 5 周 2026/08/25	Lecture #9: Diagnostic Reasoning (I) 第 9 讲：诊断推理（上）	Prof Daniel Capurro 丹尼尔·卡普罗教授
	Lecture #10: Diagnostic Reasoning (II) 第 10 讲：诊断推理（下）	Prof Daniel Capurro 丹尼尔·卡普罗教授
Week-6 01/09/26 第 6 周 2026/09/01	Lecture #11: Defining the Clinical Task Lecture #12: Study Design 第 11 讲：定义临床任务 第 12 讲：研究设计	Prof Daniel Capurro Dr. Vlada Rozova 丹尼尔·卡普罗教授 弗拉达·罗佐娃博士
Week-7 08/09/26 第 7 周 26/09/08	Lecture #13: Clinical Accuracy Metrics Lecture #14: AI Lifecycle 第 13 讲：临床准确性指标 第 14 讲：人工智能生命周期	Dr. Vlada Rozova Dr. Vlada Rozova
Week-8 15/09/26 第 8 周 15/09/26	Lecture #15: Implementation and clinical workflows 第 15 讲：实施与临床工作流程	Prof Daniel Capurro 丹尼尔·卡普罗教授
MODULE 3: Main techniques MODULE 3: Main techniques 模块 3：主要技术 模块 3：主要技术		

Week-8 15/09/26 第 8 周 2026/09/15		
Week-9 22/09/26 第 9 周 2026/09/22	Lecture #16: Deep Learning 第 16 讲：深度学习	Dr. Vlada Rozova Vlada Rozova 博士
Week-10 06/10/26 第 10 周 2026/10/06	Lecture #19: AI Agents Lecture #20: Combining Methods 第 19 讲：AI 代理 第 20 讲：方法结合	Prof Daniel Capurro Prof Daniel Capurro 丹尼尔·卡普罗教授 丹尼尔·卡普罗教授

Table 1: MODULE 4: Implementation & Legal aspects

表 1：模块 4：实施与法律方面

Week-11 13/10/26 第 11 周 13/10/26	Lecture #21: Evidence to decision framework 第 21 讲：从证据到决策的框架	Dr. Vlada Rozova Vlada Rozova 博士
	Lecture #22: Legal/Regulatory aspects of ML Applications 第 22 讲：机器学习应用的法律/监管方面	TBD
Week-12 20/10/26 第 12 周 20/10/26	Lecture #23: Project Discussion 第 23 讲：项目讨论	Prof Daniel Capurro Daniel Capurro 教授
	Lecture #24: Project Discussion 第 24 讲：项目讨论	Prof Daniel Capurro Daniel Capurro 教授
Week-14 03/11/26 第 14 周 03/11/26	Lecture #25: Project Presentations - Specific Schedule TBD, will happen during lecture and tutorial hours 第 25 讲：项目展示 - 具体时间待定，将在讲座和辅导时间进行	All 全部